

Variation in Many-Analyst Projects: Harmful or Helpful? A Multiverse Approach to Address Different Sources of Variation

Thesis Proposal

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1 Introduction

Recent studies have shown considerable variation in the results when many researchers answer the same question using the same data, raising concerns about the reproducibility of science (e.g., Hoogeveen et al., 2023; Silberzahn et al., 2018). For example, Kowall et al. (2025) asked sixteen teams to analyse whether marital status (cohabiting married vs. never married) predicts cardiovascular disease, and their estimated effects ranged from negative to positive. The sources of this outcome variation remain largely unclear. Previous studies have attributed it to different factors such as covariate choice, model type, and target population (e.g., Hoogeveen et al., 2023; Kowall et al., 2025), but systematic quantitative evidence of the relative influence of different sources of variation is lacking. Researcher decisions possibly underlie much of the variability, as variation is visible not only in results but also in analytical and substantive choices of the analysts (Table 1).

Table 1: Different choices that were made by analysts in the study by Kowall et al. (2025)

(a) Analytical decisions about data processing and statistical models.

Missing Data Handling Methods	Type of Model
Complete cases (16)	Cox Proportional Hazards model (6)
Imputed data (3)	Logistic Regression (5)
“No information” (1)	Log-binomial model (2)
	Mixed-effects logistic regression (1)
	Discrete time mixed effect model (1)
	Generalized estimating equations (1)

(b) Substantive decisions about variable definitions and target populations.

Outcome (Dependent Variable)	Adjustment Set (Covariates)	Type of Data (Change Over Time)	Exclusion Criteria
Heart attack / stroke ever (16)	Age (12)	Strictly longitudinal (10)	Subjects married before age 18
Age at heart attack / stroke (9)	Sex (12)	Longitudinal with wave 1 as the only baseline (3)	Subjects from Israel
Heart attack / stroke since last interview (7)	Country (7)	Strictly or partly cross-sectional (6)	Persons under age 55 at recruitment
Main cause of death heart attack / stroke (5)	Smoking (6)		Subjects born after 1954
	Education (4)		
	Physical activity (4)		Change in exposure

Note: The numbers in brackets behind the decisions denote how many analysts made this choice in their analysis. The outcome and adjustment sets were composed of combinations of the variables in this table.

Research Gap Many-analyst projects such as Kowall et al. (2025) remain largely descriptive. They show that both outcome and decision variability exist, but do not systematically quantify how specific researcher decisions contribute to differences in results — which is the gap I will address in this study. This lack of systematic analysis makes it difficult to interpret whether the observed variation is harmful (for credible, reproducible science) or helpful (theoretically meaningful) (Auspurg & Brüderl, 2024), highlighting the relevance of the present study.

Research Question To what extent do different types of analysis decisions, including handling missing data, choice of statistical model and definitions of outcome variables, covariates and target population, contribute to variation in the outcomes of many-analyst projects?

Contribution of Present Study To operationalize this question, I will use the uncontrolled variation in Kowall et al. (2025) to construct a decision space, containing combinations of the analysts’ choices, and explore this using *multiverse analysis*. I will systematically map the described analytical choices and vary them in a controlled manner, while holding the dataset constant, to quantify how specific analytic decisions impact outcome variation (Steege et al., 2016). Figure 1 shows an overview of the steps in the multiverse (Simson et al., 2024). With this method, this study aims to *explain* sources of variation, rather than *describing* its existence.

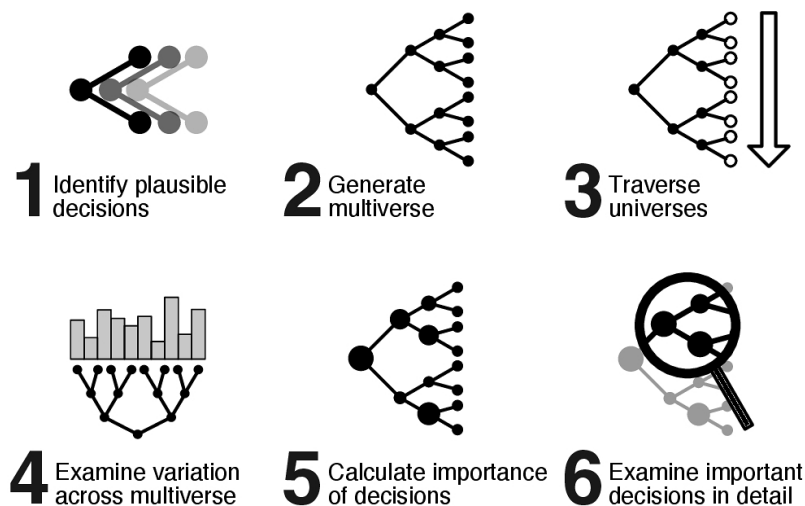


Figure 1: Comprehensive Overview of Multiverse Analysis, from Simson et al. (2024)

Expectations Consistent with previous studies, I expect substantial variation across analytical choices, but have reservations about predicting what sources will contribute most.

2 Analysis Strategy

Data Although I will not analyse the original teams’ results, I will use the same data they analysed in Kowall et al. (2025), namely the Survey of Health, Ageing and Retirement in Europe (SHARE): a freely available (after registration) panel study conducted in 28 European countries (consistent with Kowall et al. (2025), I will use wave 1-7).

Decision Space I will systematically combine the analysts’ decisions as described in Table 1. A few notes:

- As exposure, all teams but two used “cohabiting married” (CM). The exceptions also included registered partnership, and married living apart, answering different questions, I will use only CM.
- As outcome, all teams used different combinations of the four variables in Table 1.
- The adjustment sets in the original study contained 25 variables. To ensure my analysis is feasible, I will only include covariates used by at least four analysts, except the ‘no covariates’ category which will also be included. I am aware this is an arbitrary decision.
- Analysts used different exclusion criteria and data types, implying they also had different sample sizes, which Kowall et al. (2025) did not explicitly name as source of variation, but may obviously have resulted in different estimates.

Sampling and Approximation Because the number of paths increases fast (as illustrated in Figure 2), an approximation algorithm will randomly sample representative combinations. I will base my approximation on Liu et al. (2023), where research paths are sampled from the decision space, and mean effects and decision sensitivity are calculated intermediately — with sample sizes based on computational and time limits. They quantify sensitivity using a K-samples Anderson–Darling test — a non-parametric method that simultaneously evaluates whether the outcomes of k different research paths follow the same distributions, assuming that sensitive decisions lead to wider variety in distributions. Additionally, they use bootstrapping to estimate 95%-confidence intervals around the sensitivity measures.

Furthermore, the distribution of the estimates of the sampled multiverse will be compared to that of the estimates in the original study to see whether the multiverse captures the same variation as the many-analyst project.

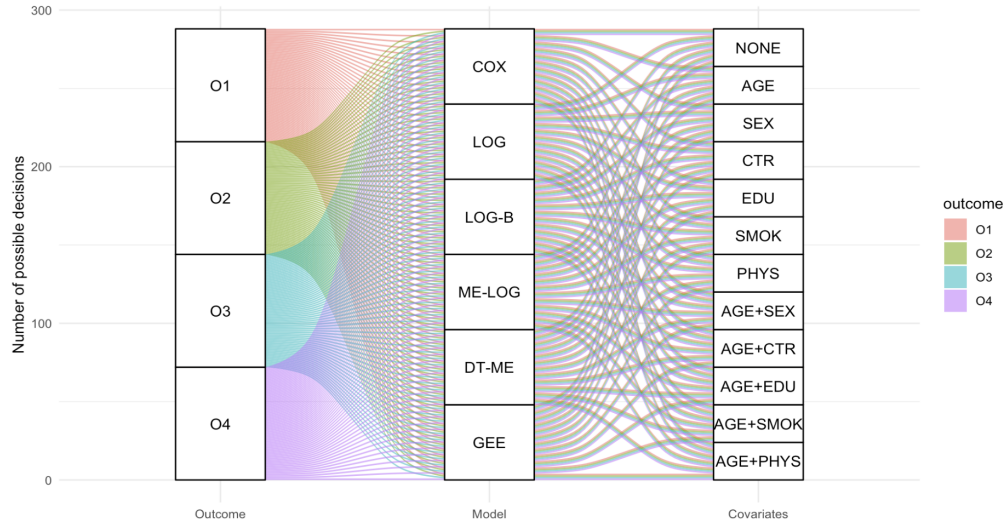


Figure 2: Simplified overview of the multiverse analysis in this study. This figure shows only a part of the multiverse; many more combinations of the outcome variables and adjustment sets are possible and other decisions will also be included. *O1-O4: outcome variables, as described in Table 1; COX: Cox Proportional Hazards model, LOG: logistic regression, LOG-B: log-binomial model, ME-LOG: mixed-effects logistic regression, DT-ME: discrete time mixed effect model, GEE: generalized estimating equations; CTR: country, EDU: education, SMOK: smoking, PHYS: physical activity.*

Software Analyses will be conducted in RStudio or Jupyter (Liu et al. (2023) provided python-code), see Table 2.

Table 2: If the analyses are performed in RStudio (version 4.4.2/4.4.3) (Team, n.d.), I wil likely use (at least) the following pack-ages:

Package	Citation
tidyverse	(Wickham et al., 2019)
ggalluvial	(Brunson, 2020)
multiverse	(Sarma et al., 2021)
specr	(Masur & Scharkow, 2020)
lme4	(Bates et al., 2015)
survival	(Therneau, 2024)
geepack	(Halekoh et al., 2006)

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